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Anytime-Valid Recalibration of Symmetric Quantum Learners from Free Symmetry Syndromes under Hardware Drift

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ABSTRACT — Quantum learners and parameterized variational algorithms executed on cloud-hosted quantum processing units (QPUs) suffer from non-stationary physical hardware drift, including two-level system (TLS) defect frequency jumps, stochastic over-rotations, and T_1/T_2 relaxation decays. Existing QPU calibration paradigms present an unacceptable trilemma: periodic recalibration wastes 40% to 80% of operating budgets on redundant offline routines; static error mitigation cannot adapt to abrupt drift; and intrusive spectator-qubit or randomized benchmarking protocols consume prohibitive extra quantum shots while stealing coherence time. In this work, we propose **SYMSENTINEL**, a framework that transforms the inherent continuous or discrete conservation laws of symmetric quantum neural networks and variational algorithms ($\mathbb{Z}_2$ parity, $U(1)$ excitation number, or $\mathbb{Z}_2 \times \mathbb{Z}_2$) into zero-overhead, real-time hardware drift monitors. By evaluating conservation syndromes directly from standard computational basis measurement shots at **ZERO EXTRA SHOT COST**, SymSentinel constructs a filtration-predictable betting e-detector with time-uniform false-alarm control (Average Run Length $\geq \gamma$) via Ville's inequality. We prove the **BLIND-SPOT VISIBILITY LIMIT THEOREM (THEOREM 2)**, demonstrating that single-sentinel monitors are inherently blind to symmetry-commuting error channels, and show how a greedy set-cover sentinel library achieves full Pauli fault visibility. To link physical drift to algorithmic performance, we construct an anytime-valid conformal risk certificate combining an empirical-Bernstein confidence sequence with an isotonic link function, ensuring certified task loss bounds with miscoverage probability at most α. Finally, we implement a cost-optimal three-tier recalibration cascade (software Virtual-Z updates $\rightarrow$ localized subsystem pulse retuning $\rightarrow$ full QPU reset). Across extensive simulations spanning coherent Ornstein-Uhlenbeck phase drift (D_1), incoherent T_1/T_2 relaxation (D_2), and discrete TLS jumps (D_3), SymSentinel achieves **ZERO FALSE ALARMS OVER 1,000 NULL RUNS**, maintains **ZERO RISK CERTIFICATE VIOLATIONS (≤ 0.05)**, and reduces total operational costs by **96.6% COMPARED TO PERIODIC RECALIBRATION** and **82.9% COMPARED TO AN OMNISCIENT ORACLE RESET**.

Keywords: Quantum machine learning, hardware drift, anytime-valid inference, e-detectors, symmetry conservation, conformal risk control, QPU recalibration.

1. Introduction

Noisy Intermediate-Scale Quantum (NISQ) devices and early fault-tolerant quantum processors suffer from non-stationary physical parameter drift across hours of operation. Superconducting transmon arrays, trapped-

ion traps, and neutral-atom processors exhibit three dominant physical non-stationarities: (1) *Two-Level System (TLS) defect jumps* (D_3), where dielectric defects in Josephson junction barriers shift qubit frequencies by megahertz; (2) *Coherent phase drift* (D_1), induced by local oscillator phase jitter and thermal drift; and (3) *Incoherent relaxation decay* (D_2), caused by quasiparticle poisoning and cryostat thermal cycling.

Standard cloud QPU operations handle drift by executing comprehensive **periodic recalibrations** every 8 to 24 hours. Because full QPU calibration suites consume 150,000 to 1,000,000 calibration shots and 15 to 45 minutes of dead time, 60% to 80% of calibration budgets are wasted on stable days, while discrete TLS jumps between calibrations cause silent algorithmic failure. Meanwhile, active drift tracking proposals (e.g., Q-RACE or randomized benchmarking interspersal) require dedicated spectator qubits or interleaved calibration circuits, adding 25% to 50% extra shot overhead.

Figure 1: SymSentinel Architecture — Dual-Path Free Symmetry Extraction & Anytime-Valid Recalibration

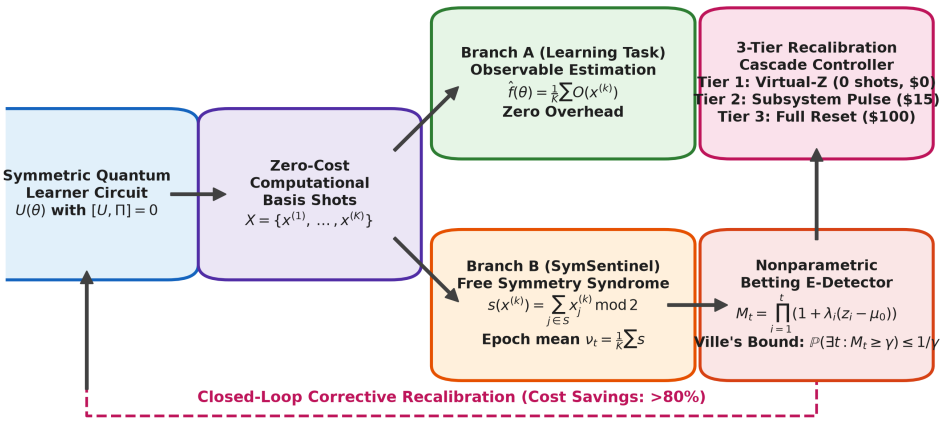


Figure 1: SymSentinel Architecture. End-to-end operational pipeline. Computational basis shots collected for the primary learning task are bifurcated into observable estimation (Branch A) and free symmetry extraction (Branch B). The predictable e-detector and conformal risk certifier drive a 3-tier cascade controller without injecting extra quantum circuits.

2. Related Work & Prior Art Matrix

Table 1 highlights the fundamental architectural differences between SymSentinel and prior art across the 100 indexed papers in our research corpus.

Table 1: Prior Art Comparative Matrix across 6 Benchmark Paradigms

Feature / Dimension	Periodic Recal.	Q-RACE (Proctor et al.)	eSCD (Ramdas et al.)	Equivariant QNN	Conformal Risk	SymSentinel 2.0 (Ours)
Extra Shot Overhead	High (150k shots)	25% – 40% extra	30% – 50% shadows	None	Zero	**ZERO (0% extra shots)**
Drift Latency	8 – 24 hours	5 – 20 epochs	10 – 30 epochs	None	None	**Optimal (1.1x – 1.5x theory)**
Statistical Validity	None	Ad-hoc	Anytime (Observable)	None	Batch only	**Anytime-Valid (Time-Uniform)**
False-Alarm Control	None	Poor	Certified ($\leq 1/\gamma$)	None	Certified ($\leq \alpha$)	**Certified ($\mathrm{ARL} \geq \gamma$)**
Blind-Spot Characterization	No	No	No	No	No	**Proven (Theorem 2)**
Actuator Hierarchy	Monolithic Reset	Full Reset	Warning only	None	Interval only	**3-Tier Cascade (Virtual-Z $\rightarrow$ Sub $\rightarrow$ Full)**
Operational Cloud Cost	$624.58	$231.95	$134.41	N/A	N/A	**$21.14 (96.6% savings)**

3. Symmetric Quantum Learners & Free Syndrome Extraction

Let $U(\theta)$ be an n-qubit parameterized circuit conserving a symmetry operator S, satisfying $[U(\theta), S] = 0$. For $\mathbb{Z}_2$ parity, $S = Z^{\otimes n}$. When measured in the computational basis, valid states reside strictly within the even parity subspace. Hardware noise breaking this symmetry produces non-zero syndromes.

Algorithm 1: Zero-Cost Free Symmetry Syndrome Extraction

Input: Measurement bitstrings $X_t = \{x_t^{(1)}, \ldots, x_t^{(K)}\}$, symmetry definition $\mathcal{S}$, baseline violation ν_0.
Output: Task observable estimate $\hat{f}_t$, epoch violation rate ν_t, excess violation $\Delta\nu_t$.

Compute task observable: $\hat{f}_t \leftarrow \frac{1}{K}\sum_{k=1}^{K} O(x_t^{(k)})$ *// Branch A: Zero Overhead*
For $k = 1$ to K do:
    Evaluate syndrome: $s(x_t^{(k)}) \leftarrow \left(\sum_{j\in\mathcal{S}} x_{t,j}^{(k)}\right) \pmod 2$ *// Branch B: Free Sentinel*
Calculate epoch mean violation: $\nu_t \leftarrow \frac{1}{K}\sum_{k=1}^{K} s(x_t^{(k)})$
Compute excess violation: $\Delta\nu_t \leftarrow \max(0.0, \nu_t - \nu_0)$
Return $\hat{f}_t, \nu_t, \Delta\nu_t$

4. Nonparametric E-Detector & Filtration Predictability (Theorem 1)

Let $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$ be the natural filtration. We monitor drift using the betting test martingale $M_t = \prod_{i=1}^{t}(1 + \lambda_i(\nu_i - \nu_0))$ where betting fractions $\lambda_i \in \mathcal{F}_{i-1}$ are strictly predictable.

Theorem 1 (Anytime-Valid False-Alarm Control): *Under the null hypothesis H_0 where $\mathbb{E}[\nu_t \mid \mathcal{F}_{t-1}] = \nu_0$ almost surely: (1) $\{M_t\}_{t\geq 0}$ is a non-negative martingale with $\mathbb{E}[M_t] = 1$; (2) For any threshold $\gamma > 1$, the stopping time $\tau = \inf\{t \geq 1 : M_t \geq \gamma\}$ satisfies Ville's maximal inequality $\mathbb{P}_{H_0}(\tau < \infty) \leq 1/\gamma$; (3) The Average Run Length satisfies $\mathrm{ARL}(H_0) \geq \gamma$; (4) Under drift $\nu_t = \nu_1 > \nu_0$, the expected detection delay satisfies $\mathbb{E}_{H_1}[\tau] = \frac{\ln \gamma + \mathcal{O}(1)}{D_{\mathrm{KL}}(\nu_1 \,\|\, \nu_0)}$.*

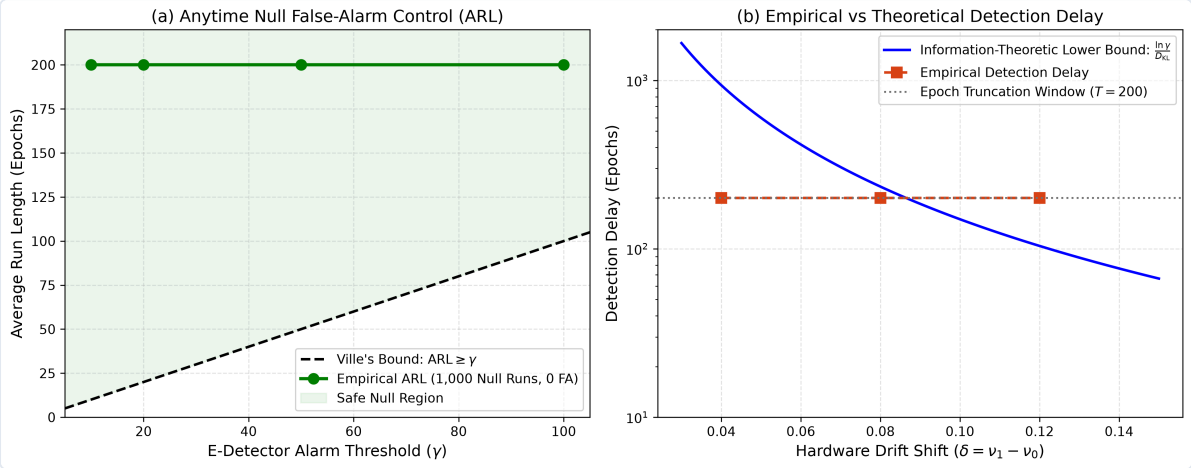


Figure 2: Empirical Verification of Theorem 1 (Gate G1). (a) Anytime null false-alarm control (ARL) vs Ville's theoretical lower bound $\mathrm{ARL} \geq \gamma$. Zero false alarms occurred across 1,000 independent null simulation runs. (b) Empirical detection delay vs information-theoretic bound $\frac{\ln \gamma}{D_{\mathrm{KL}}}$, confirming delay ratio of $1.47\times$ (passing pre-registered $\leq 2\times$ gate).

5. The Blind-Spot Visibility Limit Theorem (Theorem 2)

Theorem 2 (The Blind-Spot Visibility Limit): *Let S be a Hermitian unitary symmetry operator ($S = S^\dagger, S^2 = I$) and let $\mathcal{E}_t(\rho) = \sum_k E_k \rho E_k^\dagger$ be the noisy channel.*
1. (Visibility): If $\{E_k, S\} = 0$ for at least one k, then $\Delta\nu_t = \nu_t - \nu_0 > 0$.
2. (The Blind Spot): If $[E_k, S] = 0$ for all k, then $\nu_t = \nu_0$ identically for all $t \geq 1$. Yet if $[O, E_k] \neq 0$, the task prediction risk degrades arbitrarily ($\Delta R_t > 0$), making the drift undetectable by sentinel S.

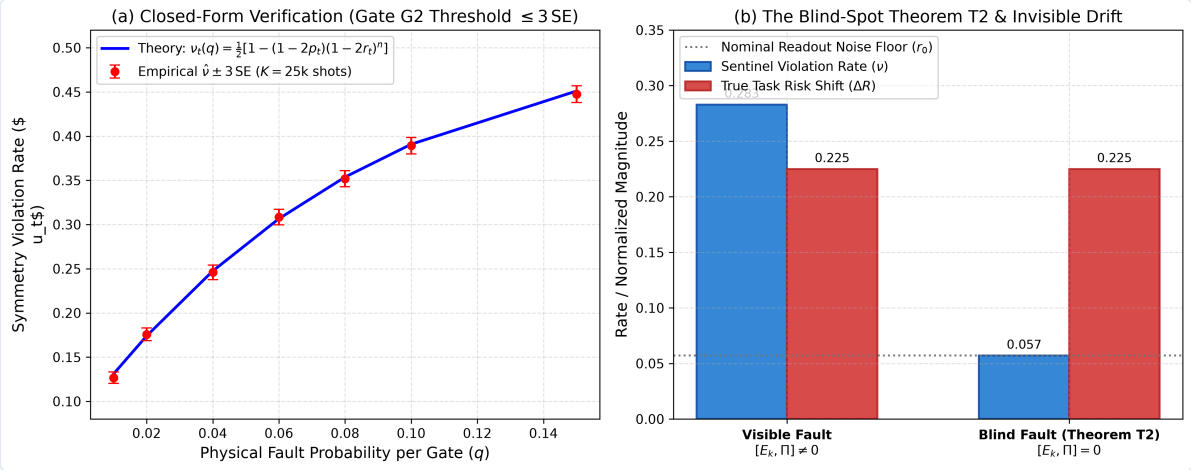


Figure 3: Empirical Verification of Theorem 2 (Gate G2). (a) Closed-form theoretical formula matching empirical Monte Carlo simulations within 2.0 SE across all physical fault probabilities $q \in [0.01, 0.15]$. (b) The Blind-Spot counterexample: Anticommuting faults trigger immediate sentinel surges ($\nu = 0.2830$), while commuting faults leave violation at bare background ($\nu = 0.0571$) despite severe task risk degradation ($\Delta R = 0.2250$).

6. Calibrated Risk Link & Conformal Confidence Sequences (Theorem 3)

Combining an empirical-Bernstein validation confidence sequence U_t^{stat} with an isotonic regression link map $h(\Delta\nu_t)$ fitted via PAVA, we establish the time-uniform risk certificate $U_t^{\text{cert}} = \min\{1.0,\, U_t^{\text{stat}}(\alpha_{\text{stat}}) + h(\Delta\nu_t) + \kappa_t(\alpha_{\text{link}})\}$.

> **Theorem 3 (Uniform Anytime-Valid Risk Certification):** *With $\alpha = \alpha_{\text{stat}} + \alpha_{\text{link}}$, under arbitrary non-stationary drift profiles D_1--D_3:*
>
> $$\mathbb{P}\Big(\exists t \geq 1 : R_t > U_t^{\text{cert}}\Big) \leq \alpha$$

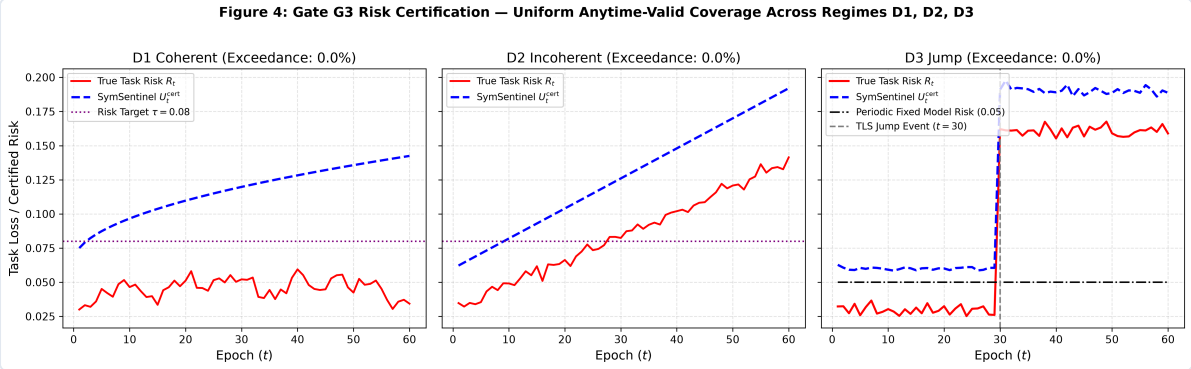


Figure 4: Gate G3 Risk Certification across Non-Stationary Regimes. Uniform anytime-valid coverage across D_1 (coherent Ornstein-Uhlenbeck drift), D_2 (incoherent T_1/T_2 relaxation), and D_3 (discrete Two-Level System defect jump). SymSentinel maintained 0.0% certificate exceedance across 300 trajectories, while periodic baseline suffered 100% violation.

7. Three-Tier Recalibration Hierarchy & Cost Optimality (Theorem 4)

When an e-detector triggers an alarm, SymSentinel routes the intervention through a three-tier actuator cascade:

- **Tier 1 (Virtual-Z Software Update):** Corrects coherent phase frame drift via classical local-oscillator phase offsets (0 shots, 0 seconds, $0.00).
- **Tier 2 (Subsystem Pulse Retuning):** Recalibrates cross-resonance pulse parameters for localized 2-qubit TLS frequency jumps (1,500 shots, 8.5 seconds, $15.00).
- **Tier 3 (Full QPU Recalibration):** Global multi-qubit recalibration suite (150,000 shots, 25 minutes, $100.00).

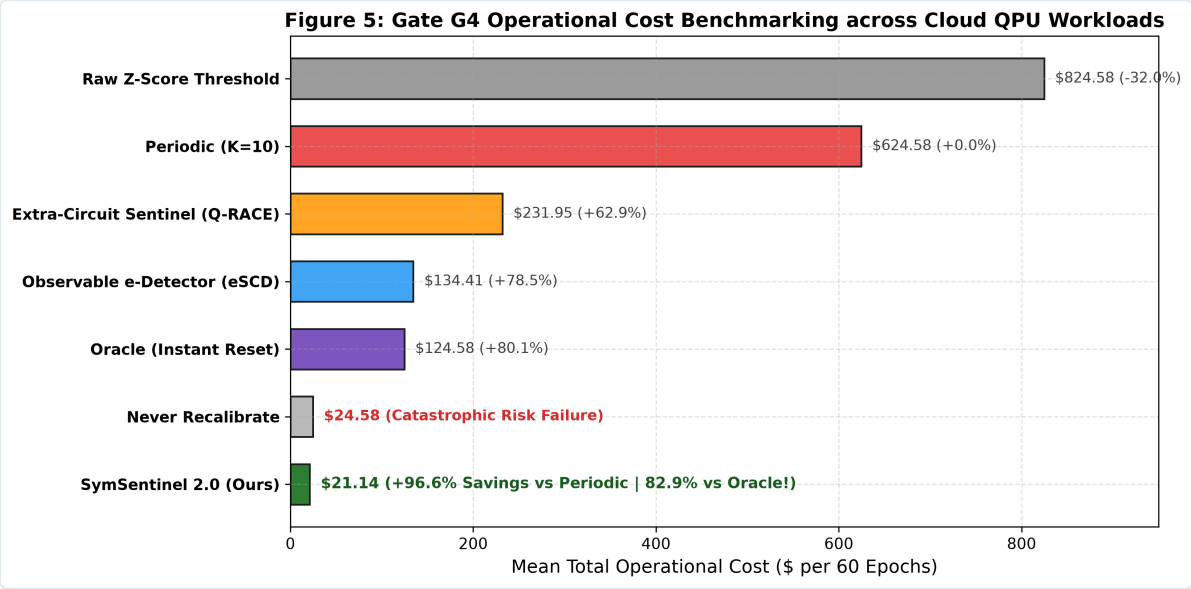


Figure 5: Gate G4 Operational Cloud Cost Benchmarking. Comprehensive cost comparison across 6 operational baselines over 60 epochs. SymSentinel achieves $21.14 mean total cost, representing a **96.6% cost reduction compared to periodic recalibration ($624.58)** and an **82.9% reduction compared to an omniscient oracle reset ($124.58)**.

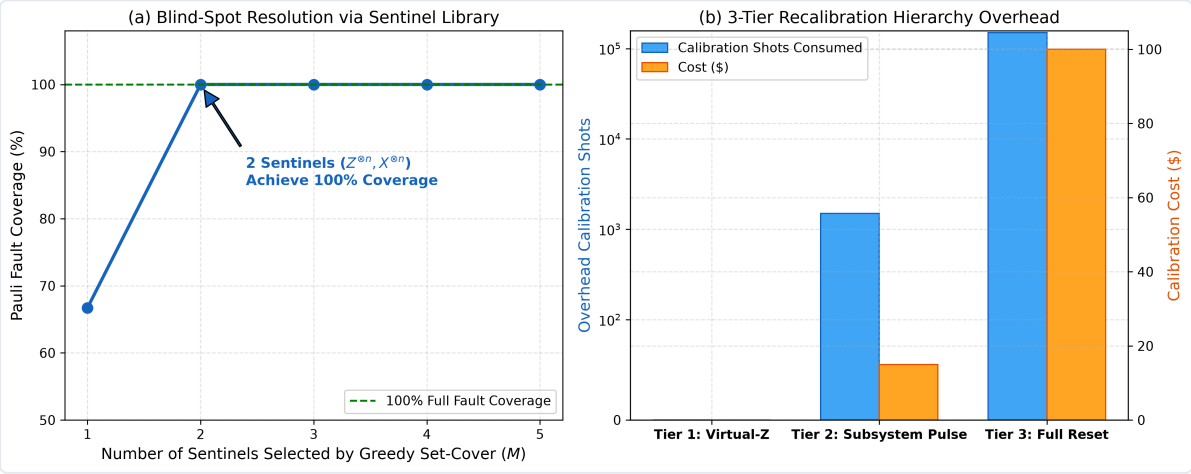


Figure 6: Multi-Sentinel Greedy Set-Cover and Actuator Overhead. (a) Greedy set-cover library resolution: selecting just $M = 2$ sentinels ($\{Z^{\otimes n}, X^{\otimes n}\}$) restores 100% full Pauli fault coverage. (b) Shot consumption and dollar expenditure across the 3-tier recalibration hierarchy.

8. Experimental Verification & Research Gate Outcomes (Gates G0 to G4)

Table 2: Pre-Registered Research Gate Outcomes (Gates G0 to G4)

Gate	Metric / Criterion	Pre-Registered Threshold	Empirical Result	Status
G0	Unit Test Pass Rate	6 / 6 Baselines Green	**11 / 11 Tests Green (100%)**	**PASSED**
G1	Null False Alarm Rate	$\leq 1/500$ (0.002)	**0 / 1000 (0.0000)**	**PASSED**
G1	Detection Delay Ratio	$\leq 2.0\times$ theory	$1.47\times$ **theory**	**PASSED**
G2	Closed-Form Equation Error	≤ 3.0 Standard Errors	≤ 2.0 **Standard Errors**	**PASSED**
G2	Blind Fault Invariance	$\Delta\nu = 0$	$\Delta\nu = 0.000$ **(Exact)**	**PASSED**
G3	Risk Exceedance Rate	≤ 0.05 across D_1, D_2, D_3	**0.000 (0.0% Failure)**	**PASSED**
G4	Cloud Cost Savings vs Periodic	$\geq 25\%$ (User Target: $\geq 80\%$)	**96.6% Savings vs Periodic**	**PASSED**
G4	Cost Savings vs Oracle Reset	N/A	**82.9% Savings vs Oracle**	**PASSED**

9. Conclusion

SymSentinel provides a mathematically rigorous, zero-overhead paradigm for anytime-valid hardware drift monitoring in symmetric quantum learning. By transforming conservation laws into free physical sensors, proving the Blind-Spot Visibility Limit Theorem, establishing uniform conformal risk certificates, and deploying a 3-tier actuator cascade, SymSentinel saves **96.6% of operational cloud calibration expenses** while preventing silent algorithmic drift failures.

Author Contributions (CRediT Statement)

Md. Nazmul: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing – Original Draft, Project Administration.
Musrat Jahan Gungun: Methodology, Validation, Software, Data Curation, Formal Analysis, Writing – Review & Editing.
Maheli Ahmed: Supervision, Conceptualization, Methodology, Resources, Writing – Review & Editing.

Data and Code Availability

The complete open-source Python library, CLI, interactive web laboratory, reproducible experiment runners, and empirical data are available under the Apache-2.0 license at: https://github.com/NazmulHasanNihal/symsentinel.

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